

ETC3550/ETC5550

Applied forecasting

Week 1: Introduction to forecasting & R



Contact details

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Tutors

- **Lachlan Macquarie**
- Sam Ferraro
- Nuwani Palihawadana
- Vincent Su

Brief bio

- Professor of Statistics, Monash University
- Co-author of most popular forecasting textbook in the world
- Developer of most popular forecasting software in the world

How my forecasting methodology is used:

- Pharmaceutical Benefits Scheme
- Electricity demand
- Australian tourism demand
- Ageing population
- COVID-19 cases

CASE STUDY 1: Paperware company

Problem: Want forecasts of each of hundreds of items. Series can be stationary, trended or seasonal. They currently have a large forecasting program written in-house but it doesn't seem to produce sensible forecasts. They want me to fix it.

Additional information

- Program written in COBOL making numerical calculations limited. It is not possible to do any optimisation.
- Their programmer has little experience in numerical computing.
- They employ no statisticians and want the program to produce forecasts automatically.



CASE STUDY 1: Paperware company

Methods currently used

- A** 12 month average
- C** 6 month average
- E** straight line regression over last 12 months
- G** straight line regression over last 6 months
- H** average slope between last year's and this year's values. (Equivalent to differencing at lag 12 and taking mean.)
- I** Same as H except over 6 months.
- K** I couldn't understand the explanation.

CASE STUDY 2: PBS



CASE STUDY 2: PBS

The Pharmaceutical Benefits Scheme (PBS) is the Australian government drugs subsidy scheme.

- Many drugs bought from pharmacies are subsidised to allow more equitable access to modern drugs.
- The cost to government is determined by the number and types of drugs purchased. Currently nearly 1% of GDP.
- The total cost is budgeted based on forecasts of drug usage.

CASE STUDY 2: PBS



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This Bulletin: **Wed, May 30 2001 6:22 PM AEST**

POLITICS

Opp demands drug price restriction after PBS budget blow-out

The Federal Opposition has called for tighter controls on drug prices after the Pharmaceutical Benefits Scheme (PBS) budget blew out by almost \$800 million.

The money was spent on two new drugs including the controversial anti-smoking aid Zyban, which dropped

the Public Record

For full election coverage

FEATURES

the Public Record

Federal Election 2001

For a fresh perspective on the federal election, reach into ABC Online's campaign weblog, [The Poll Vault](#).

CASE STUDY 2: PBS

- In 2001: \$4.5 billion budget, under-forecasted by \$800 million.
- Thousands of products. Seasonal demand.
- Subject to covert marketing, volatile products, uncontrollable expenditure.
- Although monthly data available for 10 years, data are aggregated to annual values, and only the first three years are used in estimating the forecasts.
- All forecasts being done with the FORECAST function in MS-Excel!

CASE STUDY 3: Car fleet company

Client: One of Australia's largest car fleet companies

Problem: how to forecast resale value of vehicles? How should this affect leasing and sales policies?

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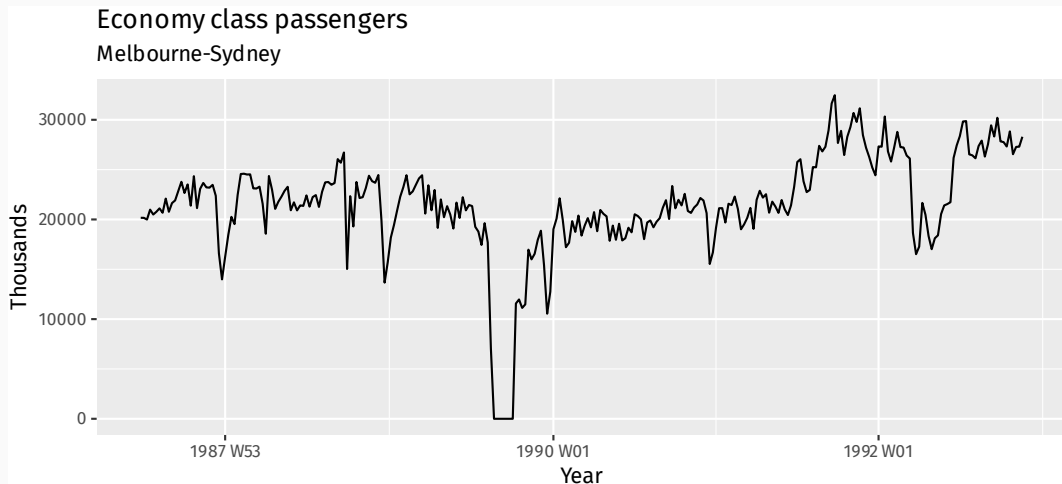
Additional information

- They can provide a large amount of data on previous vehicles and their eventual resale values.
- The resale values are currently estimated by a group of specialists. They see me as a threat and do not cooperate.

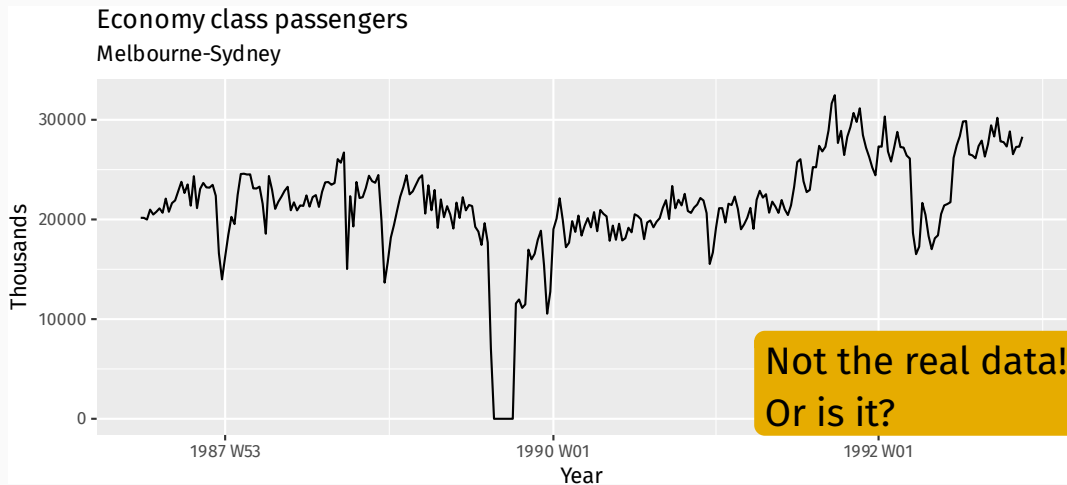
CASE STUDY 4: Airline



CASE STUDY 4: Airline



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CASE STUDY 4: Airline

Problem: how to forecast passenger traffic on major routes?

Additional information

- They can provide a large amount of data on previous routes.
- Traffic is affected by school holidays, special events such as the Grand Prix, advertising campaigns, competition behaviour, etc.
- They have a highly capable team of people who are able to do most of the computing.

Unit objectives

- 1 To obtain an understanding of common statistical methods used in business and economic forecasting.
- 2 To develop the computer skills required to forecast business and economic time series data;
- 3 To gain insights into the problems of implementing and operating large scale forecasting systems for use in business.

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Teaching and learning approach

- Approximately one hour of pre-recorded online videos each week
- One hour online lecture each Monday
- One hour in-person workshop each Tuesday
- One hour in-person tutorial on Wednesday or Thursday

Key reference

Hyndman, R. J. & Athanasopoulos, G. (2021) *Forecasting: principles and practice*, 3rd edition

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[OTexts.com/fpp3/](https://otexts.com/fpp3/)

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- Free and online
- Data sets in associated R packages
- R code for examples
- Embedded online lectures

Outline

Week	Topic	Chapter
1	Introduction to forecasting and R	1
2	Time series graphics	2
3	Time series decomposition	3
4	Simple forecasting methods	5
5	Accuracy evaluation	5
6–7	Exponential smoothing	8
8–10	ARIMA models	9
11	Multiple regression and forecasting	7
12	Dynamic regression	10

Assessment

Task	Due Date	Value
Forecasting Competition	Mon 3 Aug	2%
Weekly Quizzes	in workshop, weeks 2–11	8%
Assignment 1	Mon 24 Aug	6%
Assignment 2	Mon 14 Sep	6%
Assignment 3	Mon 5 Oct	6%
Retail Project	Fri 23 Oct	12%
Final Exam	Official exam period	60%

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Final Exam	Official exam period	60%

- Need at least 45% for exam, and 50% for total.
- **ETC5550 students:** Extra project and exam questions.

af.numbat.space

- Includes all course materials
- Links for assignment submissions
- Link to discussion forum.

Please don't send emails. Use the forum.



- The IIF provides a prize to the top student in this subject each year.
- US\$100 plus one year membership.



Available for download from CRAN:

<https://cloud.r-project.org/>



Available for download from Posit:

<https://posit.co/download/rstudio-desktop/>

Main packages



Main packages

```
# Install required packages (do once)  
install.packages(c("tidyverse", "fpp3", "GGally), dependencies = TRUE)
```

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```

```
# At the start of each session  
library(fpp3)
```

Exercises Week 1

- Make sure you are familiar with R, RStudio and the tidyverse packages.
- Do the first five modules of `starttr.numbat.space`.
- Forecasting competition

Forecasting competition: forecast the following series

- 1 Google closing stock price in \$US on 17 Aug
- 2 The difference in runs scored between Australia and Bangladesh in the second cricket test to be held 22-26 Aug, Mackay. Give result as (Australia total minus Bangladesh total)
- 3 Maximum temperature in Celsius at Melbourne airport on 13 Sep
- 4 Google closing stock price in \$US on 5 Oct
- 5 The seasonally adjusted estimate of total employment for Sep 2026. ABS CAT 6202, to be released on 15 Oct

Due Monday 3 August

For each of these, give a point forecast and an 80% prediction interval.

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- 2 The difference in runs scored between Australia and Bangladesh in the second cricket test to be held 22-26 Aug, Mackay. Give result as (Australia total minus Bangladesh total)
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Prize: \$50 cash prize

Forecasting competition: scoring

Y = actual, F = point forecast, $[L, U]$ = prediction interval

Point forecasts:

$$\text{Absolute Error} = |Y - F|$$

- Rank results for all students in class
- Add ranks across all five items

Prediction intervals:

$$\text{Interval Score} = (U - L) + 10(L - Y)_+ + 10(Y - U)_+$$

- $u_+ = \max(u, 0)$
- Rank results for all students
- Add ranks across all five items

tsibble objects

```
global_economy
```

```
# A tsibble: 15,150 x 6 [1Y]
```

```
# Key:      Country [263]
```

	Year	Country	GDP	Imports	Exports	Population
	<dbl>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>
1	1960	Afghanistan	5377777811.	7.02	4.13	8996351
2	1961	Afghanistan	548888896.	8.10	4.45	9166764
3	1962	Afghanistan	546666678.	9.35	4.88	9345868
4	1963	Afghanistan	751111191.	16.9	9.17	9533954
5	1964	Afghanistan	800000044.	18.1	8.89	9731361
6	1965	Afghanistan	1006666638.	21.4	11.3	9938414
7	1966	Afghanistan	1399999967.	18.6	8.57	10152331
8	1967	Afghanistan	1673333418.	14.2	6.77	10372630
9	1968	Afghanistan	1373333367.	15.2	8.90	10604346
10	1969	Afghanistan	1408888922.	15.0	10.1	10854428

```
# i 15,140 more rows
```

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tsibble objects

```
tourism
```

```
# A tsibble: 24,320 x 5 [1Q]
# Key:           Region, State, Purpose [304]
  Quarter Region  State Purpose  Trips
   <qtr> <chr>    <chr> <chr>    <dbl>
1 1998 Q1 Adelaide SA      Business 135.
2 1998 Q2 Adelaide SA      Business 110.
3 1998 Q3 Adelaide SA      Business 166.
4 1998 Q4 Adelaide SA      Business 127.
5 1999 Q1 Adelaide SA      Business 137.
6 1999 Q2 Adelaide SA      Business 200.
7 1999 Q3 Adelaide SA      Business 169.
8 1999 Q4 Adelaide SA      Business 134.
9 2000 Q1 Adelaide SA      Business 154.
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Domestic visitor
nights in
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tsibble objects

- A tsibble allows storage and manipulation of multiple time series in R.
- It contains:
 - ▶ An index: time information about the observation
 - ▶ Measured variable(s): numbers of interest
 - ▶ Key variable(s): optional unique identifiers for each series
- It works with tidyverse functions.

The tsibble index

Time index variables can be created with these functions:

Frequency	Function
------------------	-----------------

Annual	<code>start:end</code>
Quarterly	<code>yearquarter()</code>
Monthly	<code>yearmonth()</code>
Weekly	<code>yearweek()</code>
Daily	<code>as_date(), ymd()</code>
Sub-daily	<code>as_datetime()</code>